

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458311.76541 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.08 +/- 0.054
Transit depth (Rp/Rs)^2	0.64 +/- 0.87 [%]
Orbital Inclination (inc)	88.78 +/- 1.83
Airmass coefficient 1 (a1)	7.0 +/- 4.23
Airmass coefficient 2 (a2)	-1.76 +/- 0.82
Scatter in the residuals of the lightcurve fit is	60.04 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	3.0
Transit Duration (day)	0.09 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.08 $\pm$ 0.054 vs 0.1356 (deviation 1.03 σ)
Duration fit vs published	0.09 d vs 0.1129 d (deviation 0.67 σ)
Mid-time O-C	-1.0 min
Depth SNR	1.5
Residual scatter	60.04 %

Light curve

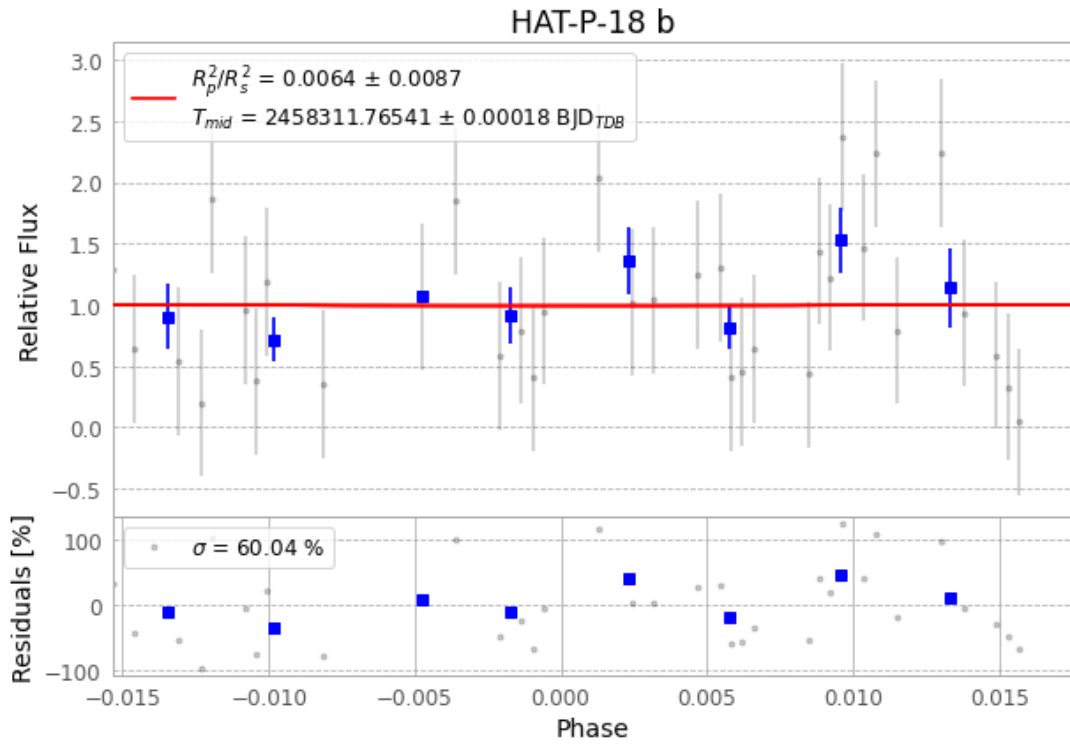


Figure 1 — FinalLightCurve_HAT-P-18 b_2018-07-11.png (SHA-256 53e96c897d7aab24...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 473], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f49891b6d3e77d1b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), HATP-18180712041512.FITS → HATP-18180712083612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-180712.log (7713ad7ba107...), FinalLightCurve_HAT-P-18 b_2018-07-11.png (53e96c897d7a...), FinalLightCurve_HAT-P-18 b_2018-07-11.csv (9c3e8e70325a...)

Compute resources

Wall time 153.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:01Z.